Enter the start and end date/time of your exam:
Sign the honor code pledge below before submitting your exam.

Start Date & Time: _____

End Date & Time: _____

ASEN 5090 Introduction to GNSS

FINAL EXAM

December 5, 2025 6PM – December 10, 2025 10PM (MST)

Format – There are two problems (each with multiple parts) worth a total of 100 points.

Submissions will include tables to fill out on the exam sheet and additional pages (handwritten or typed) answering questions, showing your work and all code used for calculations. Submission to Gradescope will be a single pdf document with pages assigned to the correct problem. Answers without supporting work will not receive credit.

Logistics – The exam period will begin on Friday, Dec 5, 2025 at 6PM and close on Wednesday, Dec 10, 2025 at 10PM (MST). If you have a question about the exam you may email or text (303-884-1297 include your name) the instructor who will respond promptly to questions on Monday-Wednesday between 10-11am.
Once you download the exam from Gradescope, you have up to 24 hours to complete the exam, scan, and upload a pdf of your solutions to Gradescope. However, all solutions must be submitted by the end of the exam period. If you run into difficulty submitting your exam on Gradescope, you may email your solution file to the instructor – but you will still be subject to the 24-hr and end of exam period limits.

Open Book – During the exam, students may use textbooks, class notes, and other references recommended for this course or that you collected/assembled prior to the date of the exam. The only internet site you may access once you begin the exam is our Canvas page. You may use a calculator, any computer codes you have already developed for the class in solving the problems, and new code you write for this exam; however, extensive coding beyond what you've already done for prior homework assignments is not expected.

During the entire exam period (6PM Dec 5 to 10PM Dec 10) you may NOT work with, consult, collaborate, or discuss the exam with ANYONE except the instructor. Once you have downloaded the exam you may NOT access or search the web, online forums, or resources other than those listed above. Any violation of these exam conditions will result in a grade of F in the course and an honor code violation report will be filed with the honor code office.

If doing calculations by hand, please assume that the speed of light, $c = 3 \times 10^8$ m/s, $\mu_E = 398600.5$ km^3/s^2. If you are using existing code, you may use these values or other established values - make sure to clearly identify the values used in your calculations.

On my honor, as a University of Colorado Boulder student, I have followed all of the above instructions and have neither given nor received unauthorized assistance on this work.

Solutions

Signature Date

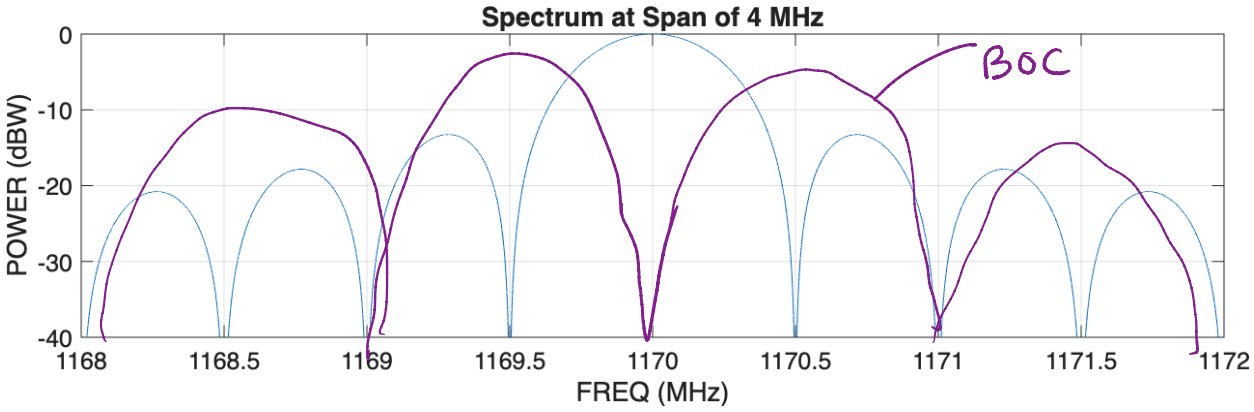
Printed name

PROBLEM 1. (50 points, parts a & b)

Figure 1 illustrates the spectrum for a proposed new navigation system. The signals are modulated with a maximal length PRN code generated with a 12 stage shift register. The satellites for this new system are in low Earth orbit at an orbital altitude of 850 km. Assume a spherical Earth with radius 6378 km.

a) Determine the carrier frequency, carrier wavelength, code chipping rate, and length of the entire PRN code. **Fill in the values in the table below** and show/explain your calculations in the space provided or on additional sheets.
In Figure 1, add a sketch of the signal spectrum that would be observed if the new system combines the code shown with a square wave to produce a BOC(1,1) signal.

	Value	Units
Carrier frequency	1170	MHz
Carrier wavelength	0.256	m
Code chipping rate	0.5	MHz
Full code length or duration	4095 bits or 8.19 ms	



Carrier frequency determined from center of spectrum = 1170 MHz
Carrier wavelength is c/fcarrier = 3e8/1.17e9 = 0.2564 or 0.26 or 0.256
Code chipping rate from lobe width is 0.5 MHz
Code length = 2^12-1 = 4095 bits, chipped at 0.5 MHz = 8.19 ms

b) Consider two receivers on the Earth surface, "Z" is directly below the transmitting satellite and "H" sees the satellite just rising along the horizon. Use the information given in the problem statement and figure in part (a), and the values given in the table below to compute results for each of the remaining table entries. **Fill in the values in the table below** and show/explain your calculations in the space provided or on additional sheets. There is no need to consider time-of-flight adjustments to the range or any other complicating factors for these calculations.

	Observer Z (zenith)	Observer H (horizon)	Units
Satellite elevation	90	0	deg
Range	850	3400.75	km
Doppler Shift	0	25,556	Hz
Space Loss	152	164	dB
C/No (*given*)	50 dB-Hz	37 dB-Hz	dB-Hz
Code Tracking sigma (*given for Z*)	0.1 m	0.45	m

Range Z = $r_{sat} - r_E = 850$ km

Range H: $r_{sat}^2 - r_E^2 = R_H^2$

Ignore motion of Earth

Doppler at zenith = 0

Doppler at el=0 $\sim \frac{V_{GPS}}{C} \cdot \frac{R_E}{R_S} \cdot f_{carrier} =$

$\frac{R_E}{C}\sqrt{\frac{\mu}{R_S^3}} \cdot f_{carrier} = 25{,}556$ Hz

Space Loss = $20\log_{10}\left(\frac{4\pi R}{\lambda}\right)$ ← Using R in km, λ in m gives 92 dB

Code tracking sig = $C_{chip}\sqrt{\frac{d}{4T\ C/N_0}}$

$\frac{\sigma_H}{\sigma_Z} = \sqrt{\frac{C/N_{0Z}}{C/N_{0H}}} = \sqrt{\frac{10^5}{10^{3.7}}}$

$\sigma_H = \sigma_Z \cdot 4.46 = 0.45$ m

PROBLEM 2. (50 points, parts a-f)

Five data files are provided for September 15, 2025

1) composite broadcast ephemeris file for this date
2 & 3) .mat files with GPS observations for a single epoch (time) for a MYSTERY location and for NIST
3 & 4) rinex ascii versions of the observation files for people whose code is set up to read this in directly

Use the data provided to solve for the mystery location. Enter your ECEF position and clock bias results for each iteration (up to 5) in the table below with precision of 0.1 m. **Attach your code and answer the questions below.**

If you are not able to successfully solve for MYST, you may submit results for NIST for partial credit.

Iteration	X (m)	Y (m)	Z (m)	Clock bias (m)
Initial Guess	100.0	0.0	0.0	0.0
1	2,615,038.3	-1,198,117.7	6,384,790.6	1,140,631.2
2	2,587,501.6	-1,045,687.5	5,730,245.3	14906.4
3	2,587,384.1	-1,043,034.9	5,716,574.7	10.1
4	2,587,384.3	-1,043,034.0	5,716,569.1	3.98
5	2,587,384.3	-1,043,034.0	5,716,569.1	4.0
Final answer in LLH	Lat: (deg) 64.13880294	Lon: (deg) -21.95549509	Ht: 97.6 m	4.0

↑ height is a bit off from correct location for 10 deg mask, $\frac{2.0}{\sin el}$

a) Please use the C1C and C2W pseudoranges to compute your solution. Why do we use two different pseudoranges for positioning?

To correct for ionospheric effects

answer with better trop model is closer.

b) Choose an initial guess for the location and justify this choice.

I tried different positions near the center of Earth to see which converged to a point on the surface. R0 of [100, 0, 0] worked fine.

Both are OK here

c) Choose an elevation cutoff and justify this choice.

Initially chose 0, then tried 5 and 10 deg. 5 or 10 both work ok. 0 gives - poor results

↑ Shown

d) Choose a tropospheric model to use and a reasonable zenith delay value.

I used zenith delay of 2m with the simple 1/sin(el) mapping and also with the split dry and wet mappings given in Slide 17 in Lecture 13. The split one works for elevation masks down to 0, but the simple one doesn't work well below 5 deg.

This data set showed very dramatically that my suggestion earlier in the semester to clip the model was (unintentionally) a really bad idea - if you used this approach the post-fit residuals will be too big - multiple meters or 10s of meters, which indicates something is wrong. In grading I gave full credit for solutions using this approach, but took off for not noticing that the residuals were bad.

e) How did you determine if your position & clock solution is correct? Describe the convergence of your iteration.
Give final values for the post-fit residuals for each satellite.

converged to a small clockerr + went to same value with different IC's + tropos

PRN	Res	el
4	0.59	25 deg
11	-0.91	24.5
25	0.60	37.4
26	-0.22	37.9
28	-1.21	63
29	0.84	56
31	0.32	63

Took ~ 4 steps to converge
First two are huge.

f) What do you notice about the MYST observations compared to what you see with the NIST data in HW5 (or with the attached file)? Compute the HDOP and VDOP for the MYST solution and compare these values to the DOPs for a NIST solution. (You can compute a solution using the NIST file provided, or refer back to your NIST results for HW5.) Briefly explain why they are similar or significantly different.

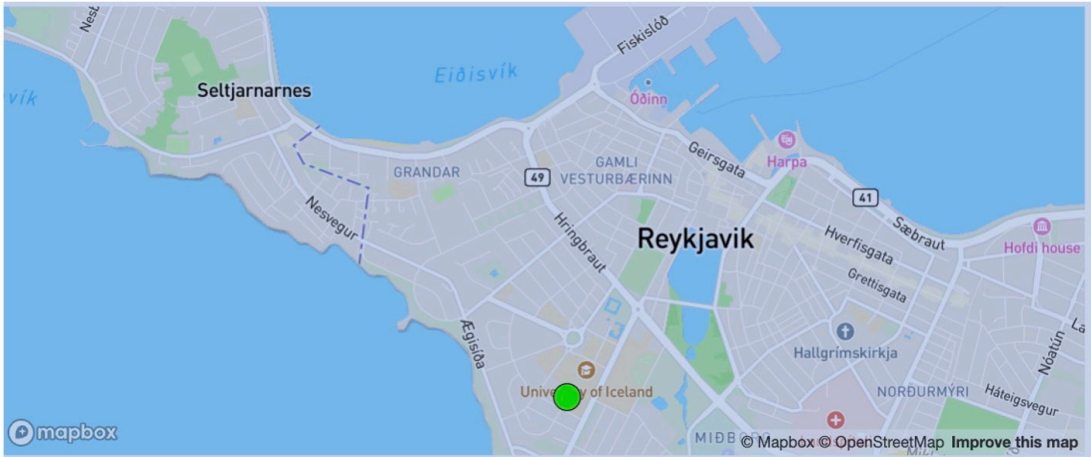
MYST has more satellites, HDOP = 1.8 VDOP = 3.9 with elev cutoff of 10 deg

with el cutoff of Ø

	HDOP	VDOP
MYST	0.70	0.95
NIST	0.86	1.06

My expectation that MYST (in Iceland) would have better HDOP than NIST and worse VDOP since it is high latitude, but it is pretty close to the GPS orbit inclination, so we actually had 5 satellites above 50 deg elevation, which ended up slightly better than NIST for this particular time.

If you picked a higher elevation mask it would cut out satellites 9 and 16 that were both below 5 deg. As you go higher on the cutoff the advantages of being closer to the pole are diminished.



Country/Region	Reykjavik, Iceland
Latitude, Longitude	64.138789, -21.955492
Elevation	93.0 (m)